



OLIST E-COMMERCE DATA ANALYSIS REPORT

(Customer Behavior, Delivery Performance & Business Insights)



1. INTRODUCTION

This project analyzes the **Olist Brazilian E-commerce dataset (2016–2018)** to understand customer behavior, delivery performance, payment patterns, and product category trends.

The dataset includes multiple interconnected tables such as:

- Orders information
- Customer details
- Payment transactions
- Product information
- Customer reviews
- Seller and logistics data



OBJECTIVE

The objective of this analysis is to:

- Understand customer purchasing behavior
- Identify factors affecting customer satisfaction
- Analyze delivery performance impact
- Study payment and product trends
- Generate actionable business insights



Business Problem

E-commerce platforms face challenges such as:

- Delayed deliveries affecting customer satisfaction
- Variation in payment behavior and spending
- Uneven product category performance
- Customer experience inconsistency

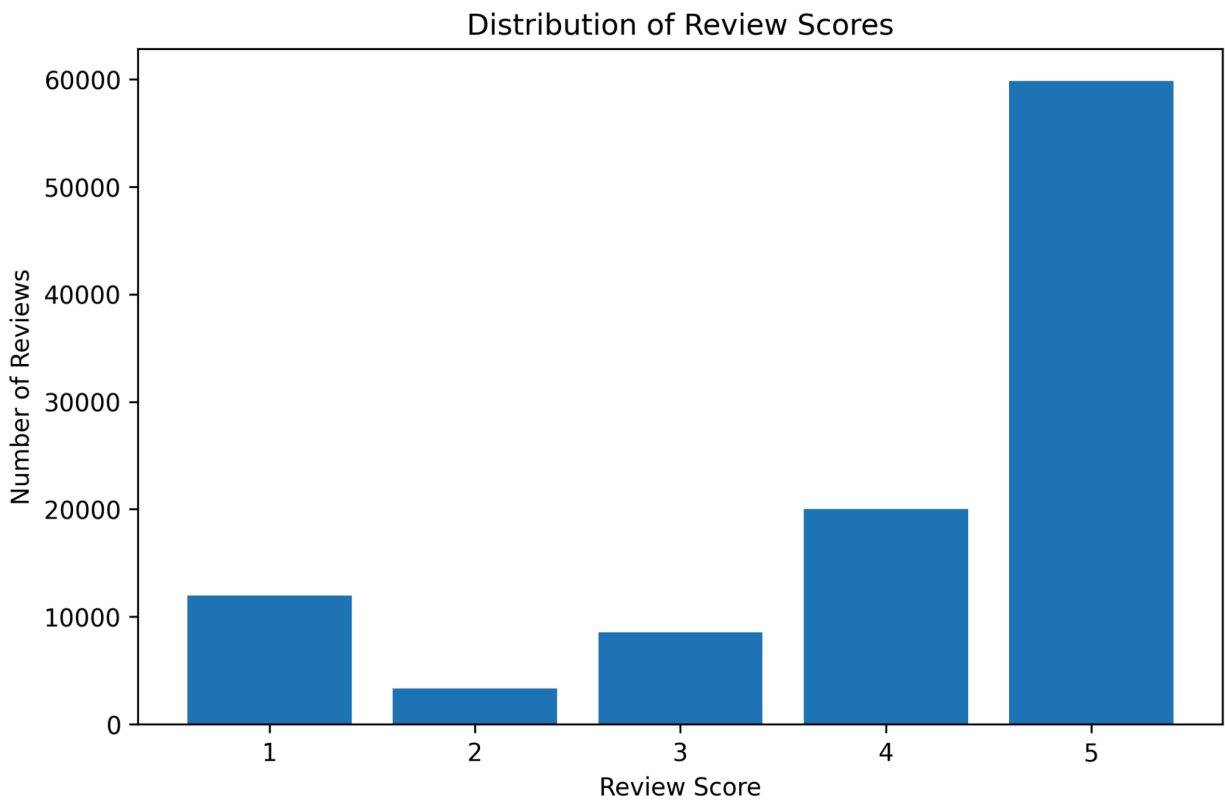
This analysis aims to identify key drivers behind these challenges.

2. EXPLORATORY DATA ANALYSIS (EDA)

2.1 Customer Behavior Overview

The dataset shows a large number of customers distributed across multiple regions in Brazil. The platform demonstrates significant transaction volume, indicating strong market presence.

★ 2.2 Review Score Distribution



Customer review scores show variation across orders, indicating inconsistent customer satisfaction levels.

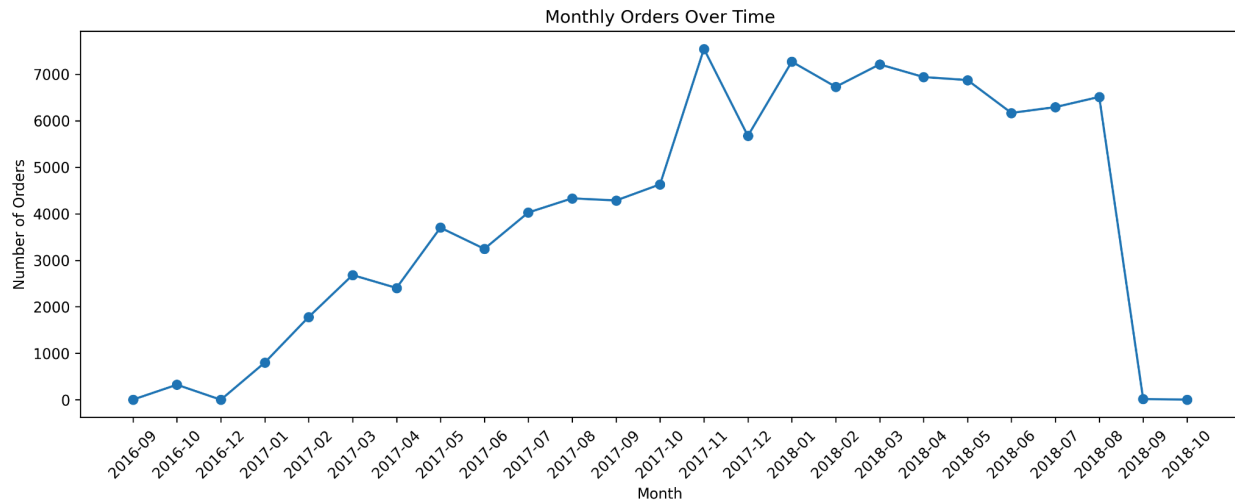
Key Insight:

- Majority of ratings are concentrated at higher and lower ends
 - This suggests polarized customer experiences
 - Some customers are highly satisfied while others face service issues
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2.3 Monthly Order Trends

Order volume analysis over time shows:

- Steady growth in transactions
- Seasonal fluctuations in demand
- Lower values in initial and final months due to partial dataset coverage



Key Insight:

The platform shows positive growth in order volume over time, indicating business expansion.

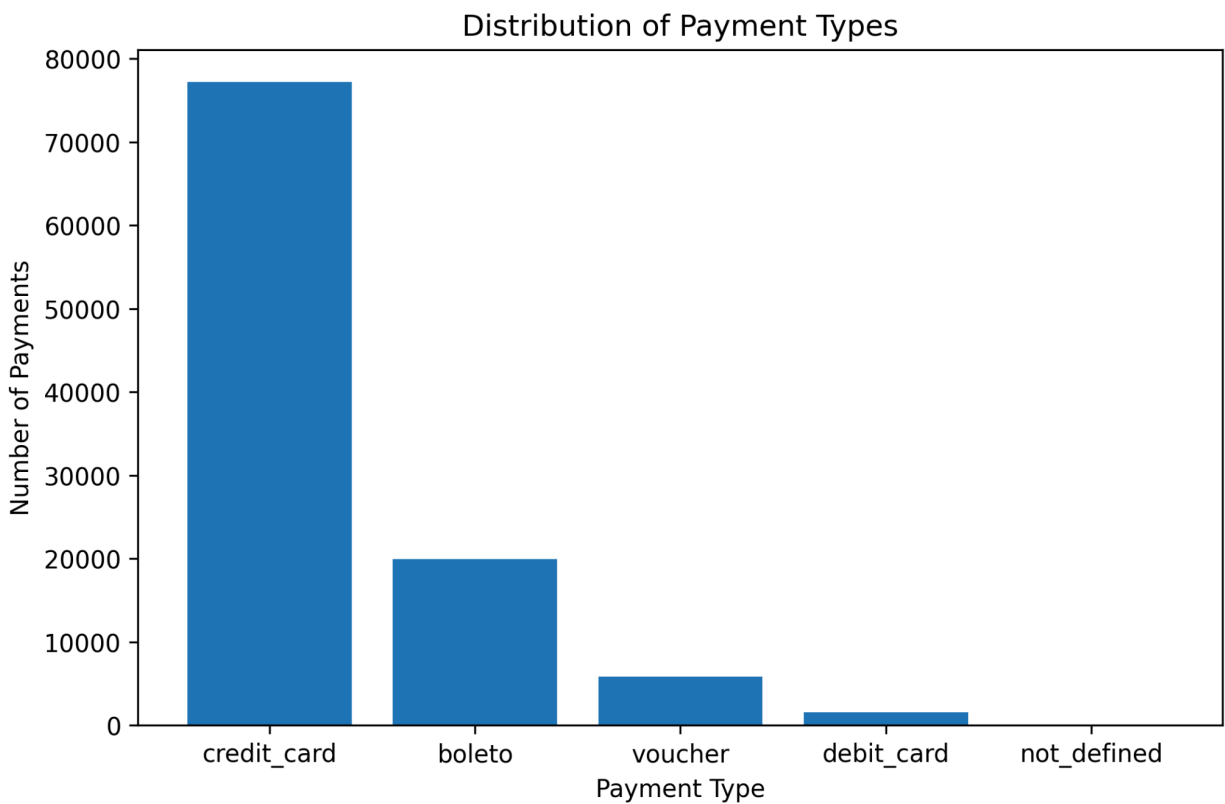
2.4 Payment Behavior Analysis

The most frequently used payment method is:

- Credit card (dominant method)

Other methods include:

- Boleto (bank slip)
- Debit card
- Voucher payments



Key Insight:

Credit card is the preferred payment method, indicating higher digital payment adoption.

3. DELIVERY & PRODUCT PERFORMANCE ANALYSIS

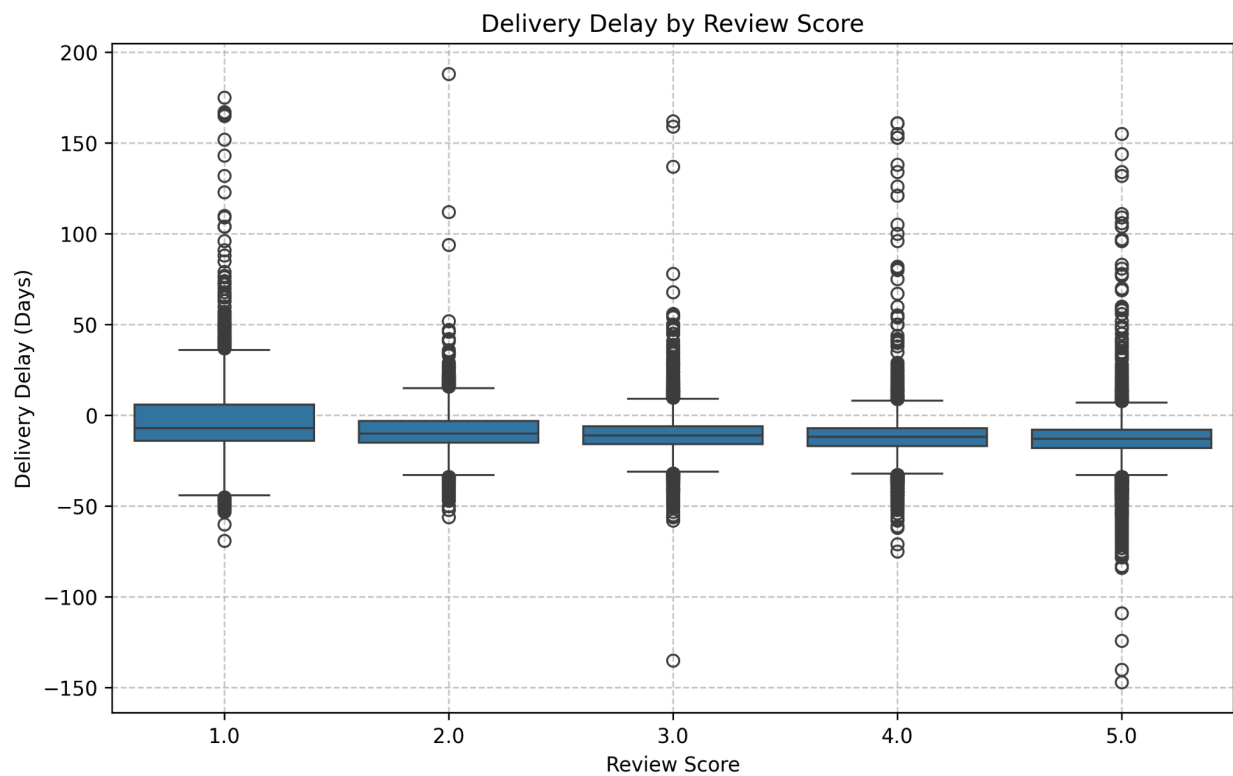
3.1 Delivery Delay Analysis

Delivery delay was calculated as:

Delivered date – Estimated delivery date

Observations:

- Positive values → delayed deliveries
- Negative values → early deliveries



Key Insight:

Negative values indicate early deliveries, which is a positive operational outcome, while positive delays indicate inefficiencies.



3.2 Impact of Delivery on Customer Ratings

A strong relationship was observed between delivery delay and customer reviews.

Statistical Insight (t-test result):

- p-value = 0.0 (highly significant)
- Delivery delays strongly affect customer ratings

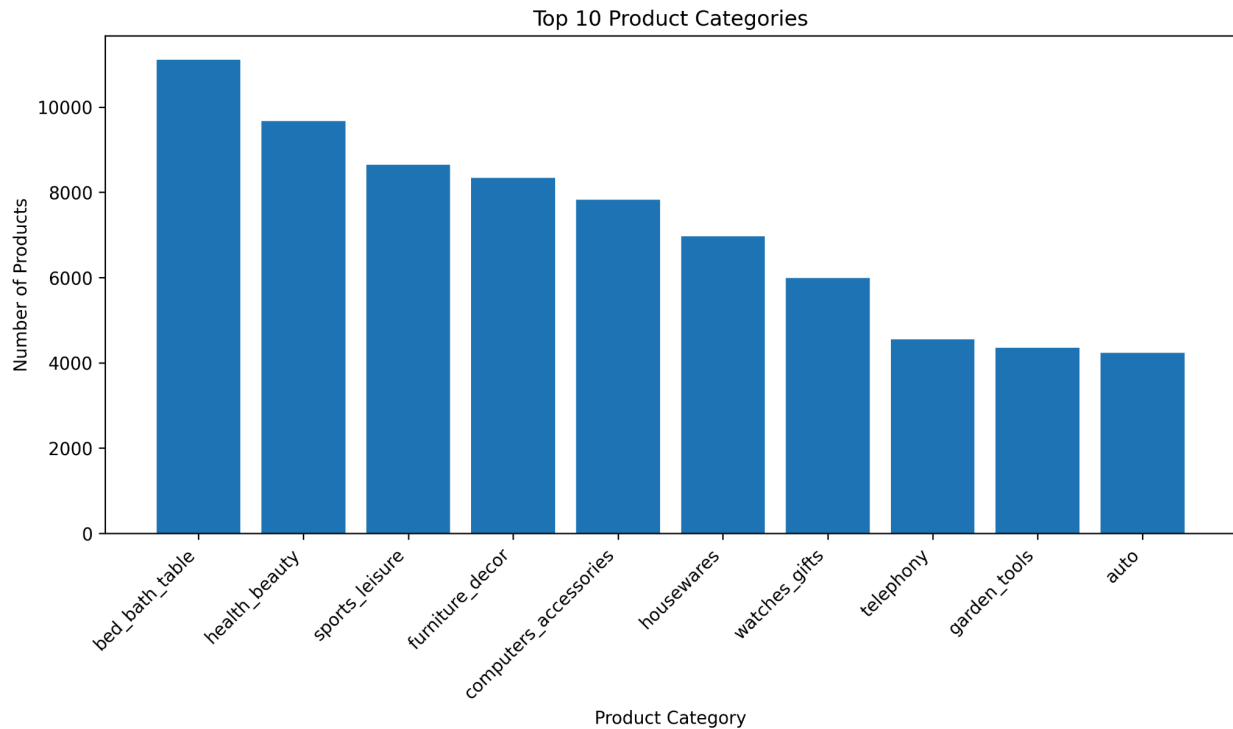
Conclusion:

Delayed deliveries significantly reduce customer satisfaction.

3.3 Product Category Performance

Top-performing product categories include:

- Bed & bath
- Health & beauty
- Furniture & decor
- Sports & leisure



Key Insight:

Certain categories dominate order volume, while others contribute less to overall demand.

3.4 Category-wise Performance Insight

Some product categories show lower average ratings despite high order volumes.

Insight:

This indicates quality or service inconsistencies within specific categories.

4. HYPOTHESIS TESTING

4.1 Hypothesis 1

Delivery delays impact customer review scores

Method:

Independent t-test comparing:

- Delayed orders
- On-time/early orders

Result:

- p-value = 0.0 (statistically significant)
- Strong difference in means observed

Conclusion:

Delivery delays significantly reduce customer satisfaction.

4.2 Hypothesis 2

Payment type affects order value

Method:

t-test between:

- Credit card users
- Boleto users

Result:

- p-value = $3.14e-25$ (highly significant)
- Credit card users spend more on average

Conclusion:

Payment method has a significant impact on spending behavior.

5. KEY INSIGHTS & BUSINESS RECOMMENDATIONS

5.1 Key Insights

1. Delivery performance is a critical driver of customer satisfaction
 2. Credit card users generate higher revenue per order
 3. Customer satisfaction varies significantly across orders
 4. Product category performance is uneven across segments
 5. Demand shows seasonal and growth trends over time
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5.2 Business Recommendations

1. Improve Delivery Operations

- Optimize logistics network
- Reduce delivery delays
- Improve estimated delivery accuracy

2. Enhance Customer Experience

- Monitor delayed orders proactively
- Improve communication with customers
- Reduce negative delivery experiences

3. Optimize Payment Strategy

- Promote credit card usage
- Offer incentives for high-value transactions
- Improve checkout experience

4. Product Strategy Improvements

- Improve low-rated categories
 - Focus on quality control in underperforming segments
 - Strengthen seller management
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5.3 Future Scope

Future analysis can focus on:

- Customer lifetime value (CLV)
 - Repeat purchase behavior
 - Customer segmentation (RFM analysis)
 - Cohort retention analysis
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6. CONCLUSION

This analysis provides a comprehensive understanding of customer behavior, delivery performance, and business dynamics within the Olist e-commerce platform.

Key findings highlight that:

- Delivery efficiency strongly influences customer satisfaction
- Payment methods impact revenue generation
- Product categories show varied performance patterns

These insights can help improve operational efficiency, customer experience, and revenue growth strategies.

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